

A Multi-Timescale Resilient Food Portfolio

Robust Optimization of Emergency Nutrition, Biomass Production, and Fail-Closed Safety

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Abstract

Emergency food planning fails when unlike quantities are treated as interchangeable: biological growth is counted as edible delivery, protein is counted as complete nutrition, an unknown safety state is replaced by an optimistic estimate, or a solver response is mistaken for proof that a safe plan exists. The deeper problem is temporal. Stored food, functioning markets, cash, humanitarian delivery, local cultivation, and controlled biomass production operate on different clocks and under different forms of authority. A useful portfolio must therefore coordinate immediate nutrition and water with later production while preserving product identity, process control, testing, regulation, population suitability, cultural acceptability, fallback capacity, and recovery. I develop a Multi-Timescale Resilient Food Portfolio (MRFP), a typed and fail-closed decision architecture that separates organism growth, gross production, rejected and lost output, released food, delivered food, and consumed nutrients. Unknown mandatory evidence contributes no released output. A lexicographic objective protects hard safety, water, clinical, and legal conditions before minimizing nutrient shortfall, common-mode dependency, inequity, recovery time, and cost. The architecture returns explicit infeasibility or computational refusal instead of manufacturing a favorable answer. The accompanying reference application was tested in successive synthetic programs. Development fixtures executed 84 policy-scenario cells, 18 ablations, and 24 adversarial fixtures. A preregistered 72-case partition retained 432 condition-case results and 27,648 candidate evaluations. A subsequent sensitivity program retained 121,608 candidate evaluations. A separately frozen mixed-integer program initially failed exact replay twice: one study differed at 1 of 384 identities, and a recovery study produced two opposite-pass timeouts. Neither was aggregated. A new precommitted, single-threaded HiGHS 1.15.1 route then reproduced three 96-result development passes and two 384-result scientific passes exactly. Its nine endpoints included favorable, null, tie-heavy, and adverse results. In particular, the model reduced false feasibility under uncertainty protection, but its first complete-nutrient formulation lost the micronutrient comparison, a fractional-allocation sensitivity was exactly null, and a fragility constraint increased cost in all 48 paired cases. A Lean 4 leaf checks 29 bounded invariants covering fail-closed solver classification, replay gates, endpoint accounting, direction, evidence-history preservation, seal state, abstract release, and an inventory balance certificate. This formal layer does not prove solver optimality over exact reals or empirical food-system performance. The result is a reproducible research framework for discovering when a candidate production module helps, ties, fails, or cannot yet be judged. Existing food assistance, safe water, established nutritious supply, local governance, and qualified evidence remain primary.

Keywords: emergency nutrition; humanitarian logistics; robust optimization; food-system resilience; food safety; fail-closed decision support; single-cell protein; microalgae; edible insects; mushrooms; reproducibility

1. The problem: food arrives on several clocks

Imagine that a storm, conflict, cyberattack, or infrastructure failure isolates a community. A warehouse contains grain and oil. Some shops still have food. Water treatment is unreliable, refrigeration is failing, roads may reopen in four days, and a nearby laboratory proposes growing protein-rich biomass. These resources are not interchangeable.

Stored food may provide calories now. Markets or cash may help if trade and access still function. Humanitarian delivery may add established foods and specialized products when routes reopen. A mushroom crop cannot feed anyone before colonization and fruiting. A microbial reactor cannot contribute safe human food merely because an organism divides quickly: the facility must exist, receive qualified inputs, remain controllable, process its output, pass testing, satisfy regulatory and population-specific conditions, and distribute a food that people can safely and willingly consume.

The planning problem is therefore not how to maximize the growth rate of one organism. It is how to coordinate food, water, production, testing, logistics, and governance across several time horizons without allowing an attractive new module to displace a less glamorous but more reliable baseline. A plan can possess abundant protein and still fail because it lacks energy, essential fats, micronutrients, safe water, digestibility, acceptable preparation, clinical separation, distribution, or evidence.

This framing begins with the ordinary experience before introducing new vocabulary. People need safe, usable nutrition over time. The system must distinguish what exists, what can be delivered, what can be produced later, and what is merely promised. It must also be able to say no: no valid input, no certified computational result, no safe feasible portfolio, or no authority to act.

2. Established methods and the remaining gap

Humanitarian food planning already integrates needs assessment, food-basket construction, sourcing, transport, storage, transfer modality, cost, and operational constraints. The World Food Programme describes context-sensitive baskets designed to meet energy and nutrient needs and uses 2,100 kcal per person per day as a common full-dependence planning reference, not an individualized prescription [1]. Its nutrition strategy and Fill the Nutrient Gap program place diet quality, affordability, and local context inside the planning problem [2,8]. Children with wasting or nutritional oedema require specialized, clinically governed prevention and management rather than substitution by a generic high-protein product [3,4].

Operations research has already advanced the field. Peters and colleagues linked nutrition with procurement and delivery decisions in a nutritious humanitarian supply-chain model [5]. De Moor and colleagues compared nominal, robust, adaptive-robust, and rolling-horizon methods under uncertainty [6]. Ekici and Ozener developed an adjustable robust approach to humanitarian relief distribution [7]. MRFP therefore does not claim to invent food baskets, robust optimization, or adaptive humanitarian logistics.

Food production introduces a different dependency chain. Codex food-hygiene principles distinguish prerequisites, hazard analysis, critical limits, corrective action, verification, documentation, traceability, and recall [9]. The statement "the culture grew" is not equivalent to "safe food was released," and release is not equivalent to delivery or consumption. Safe water, sanitation, food handling, refrigeration, laboratory capacity, and qualified personnel remain inside the nutrition

system [10-12,27].

Candidate production modules also differ materially. Insect risk depends on species, feed substrate, life stage, production, processing, allergenicity, and jurisdiction [13-15]. *Arthrospira* has a genuine food history and nutritional value, but commercial-product studies also identify product-specific cyanotoxin, microbial, and heavy-metal concerns; an untested open culture cannot inherit the evidence of a controlled and tested product [16-20]. Oyster-mushroom timing and yield depend on species, spawn, substrate, preparation, temperature, contamination, and other conditions [21,22]. Single-cell protein can use different organisms and feedstocks, but reactor output still requires downstream processing, composition, digestibility, nucleic-acid, hazard, regulatory, and complete-diet accounting [23-25].

The remaining gap is a joined test architecture. Conventional assistance and candidate production need explicit time and authority layers. Each module needs a complete typed state. Unknown evidence must fail closed. Robust and adaptive policies need matched established baselines. The program must treat inability to supply safe, complete nutrition as a result rather than an optimization inconvenience.

3. Origin and contribution

The food-specific source center is my August 16, 2025 public article, *Beating Famine With Fast-Regenerating Foods*. It proposed three coordinated planning horizons: a 72-hour household or shelter bridge, a 7-14-day neighborhood layer, and a 48-90-day municipal or institutional transition. It distinguished algae, fungi, insects, and microbial protein; preserved humanitarian access; and included water, safety, local conditions, cultural choice, and dignity. These intervals are authored planning bins, not universal biological or humanitarian thresholds.

Earlier work supplies adjacent ancestry rather than the full food architecture. A 2020 emergency-production proposal considered industrial mobilization under shock. A 2024 risk essay joined cumulative hazards with intervention and adaptation. The 2025 article fixed the three-horizon food-specific proposal. The mathematical, software, statistical, and formal developments reported here were produced in 2026 and are not backdated.

The operation can be stated without its acronym:

1. State what each population needs and what safe supplies already exist.
2. State when each candidate could deliver released food, not merely when its organism could grow.
3. Charge every input: safe process water, energy, substrate, labor, testing, processing, refrigeration, transport, and storage.
4. Disqualify output whose identity, composition, process, hazard, test, regulatory, population, or traceability evidence is missing.
5. Test the remaining portfolio against multiple simultaneous failures and alternative observations.
6. Preserve fallback inventory, local choice, clinical separation, and a route back to ordinary supply.
7. Return invalidity, computational refusal, or infeasibility when the declared conditions cannot be satisfied.

I call this joined operation a **Multi-Timescale Resilient Food Portfolio**. Its novelty is the combined architecture and test program, not the invention of its inherited components.

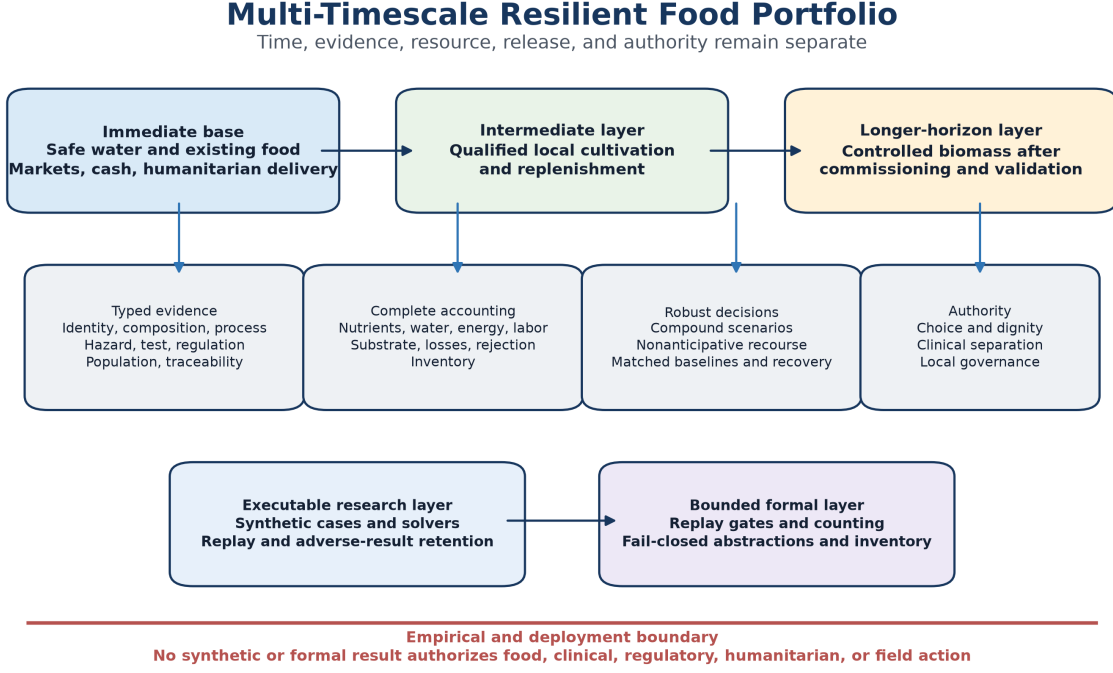


Figure 1. MRFP system context. Established food assistance and safe water remain the first-line base. Candidate production enters only through typed evidence, resource, release, and governance gates. Synthetic optimization and bounded formal verification do not cross the empirical or deployment boundary.

4. Typed system boundary

Let $g \in \mathcal{G}$ index population groups, $n \in \mathcal{N}$ index energy and nutrient dimensions, $j \in \mathcal{J}$ index food, transfer, or production modules, $r \in \mathcal{R}$ index resources, $t \in \mathcal{T}$ index decision periods, and $\omega \in \Omega$ index disruption scenarios.

The module taxonomy contains conventional stock, market transfer, humanitarian delivery, local cultivation, controlled biomass, clinical nutrition, and water/WASH capacity. Type is not a safety grade. Every physical food module must have a declared identity, intended population, jurisdiction, input specification, process envelope, measurement plan, release criteria, storage rule, and recall route.

If $P_{gt\omega}$ is the population of group g , and $R_{gnt\omega}$ is its versioned planning requirement for nutrient n , then total required amount is

$$N_{gnt\omega}^{req} = P_{gt\omega} R_{gnt\omega}. \quad (1)$$

The model does not infer a clinical diet from age or population label alone. Clinically governed products and groups remain on their own eligibility path.

Each module has a state vector that includes time, capacity, identity, preparation, nutrient composition, resource use, hazard controls, test validity, jurisdiction, population suitability,

acceptability, dependencies, recovery, provenance, and uncertainty. A partial record is not silently completed from a different organism, preparation, facility, study, or jurisdiction.

5. Time, production, release, and delivery

Let x_{jt} be commissioned capacity, K_{jtw} usable capacity, K_{jw}^0 existing capacity, and ℓ_j validated commissioning delay. Capacity cannot appear before its lag:

$$K_{jtw} \leq K_{jw}^0 + \sum_{\tau \leq t - \ell_j} x_{j\tau}. \quad (2)$$

Let q_{jtw}^{gross} be gross physical output and y_{jtw} a validated yield factor:

$$0 \leq q_{jtw}^{gross} \leq y_{jtw} K_{jtw}. \quad (3)$$

Gross biomass is not edible delivery. If L_{jtw} is processing and handling loss, $B_{jtw} \in \{0, 1\}$ is batch acceptance, and q_{jtw}^{rel} is released output, then

$$q_{jtw}^{rel} \leq (1 - L_{jtw}) q_{jtw}^{gross}, \quad q_{jtw}^{rel} \leq M_j B_{jtw}. \quad (4)$$

The first horizon receives a conservative default: newly improvised biological production contributes no released output. A pre-existing qualified facility may contribute only within its approved identity, input, process, staff, testing, storage, and distribution envelope.

6. Fail-closed evidence and hazards

For every module-period-scenario tuple, define eight binary evidence gates for identity, composition, process, hazard, test, regulation, population suitability, and traceability:

$$e^{id}, e^{comp}, e^{proc}, e^{haz}, e^{test}, e^{reg}, e^{pop}, e^{trace} \in \{0, 1\}. \quad (5)$$

The qualification variable Z_{jtw} is one only when every mandatory gate is one:

$$Z_{jtw} \leq e_{jtw}^k \quad \forall k, \quad Z_{jtw} \geq \sum_k e_{jtw}^k - (|k| - 1). \quad (6)$$

Released output is bounded by qualification:

$$q_{jtw}^{rel} \leq M_j Z_{jtw}. \quad (7)$$

An unknown mandatory gate is encoded as zero, not as an average or favorable guess. For measurable hazard h , with conservative upper bound U_{jhtw} and applicable limit θ_{jh} , acceptance also requires

$$U_{jhtw} \leq \theta_{jh} \quad \forall h \in \mathcal{H}_j. \quad (8)$$

Hard-stop hazards are not combined into a compensatory score. Extra protein cannot offset a failed toxin, pathogen, allergen, substrate, gas-safety, or regulatory condition.

7. Nutrient, resource, inventory, and dignity accounting

Let a_{jn} be the verified nutrient amount per released unit, $d_{jg} \in [0, 1]$ the population-specific usable fraction after digestibility and suitability controls, and $v_{jg\omega}$ the allocation to group g :

$$0 \leq \sum_g v_{jg\omega} \leq q_{j\omega}^{rel}. \quad (9)$$

Delivered usable nutrient is

$$D_{gnt\omega} = \sum_j a_{jn} d_{jg} v_{jg\omega}. \quad (10)$$

Shortfall remains explicit:

$$s_{gnt\omega} \geq N_{gnt\omega}^{req} - D_{gnt\omega}, \quad s_{gnt\omega} \geq 0. \quad (11)$$

The bounded satisfaction ratio is

$$\rho_{gnt\omega} = \min \left(1, \frac{D_{gnt\omega}}{N_{gnt\omega}^{req} + \varepsilon} \right). \quad (12)$$

Energy, protein quality, essential fats, and critical micronutrients remain separate dimensions. Reporting aggregate protein alone is prohibited. Vulnerable groups retain distinct requirements and exclusions.

If c_{jr} is the amount of resource r consumed per unit of gross output, f_{jr} is fixed commissioning demand, and $A_{rt\omega}$ is safely available resource, then

$$\sum_j c_{jr} q_{jt\omega}^{gross} + \sum_j f_{jr} x_{jt} \leq A_{rt\omega}. \quad (13)$$

Gross rather than released output appears in the production term because rejected batches still consume water, energy, substrate, labor, and capacity. Culture or cleaning water is not counted as drinking water. Aeration, sterilization, drying, refrigeration, temperature control, and gas handling are not free.

For stored commodity k , with deliveries $u_{kt\omega}$, consumption $c_{kt\omega}^{food}$, rejection $r_{kt\omega}$, and storage loss $\lambda_{kt\omega}$, inventory obeys

$$I_{k,t+1,\omega} = I_{kt\omega} + u_{kt\omega} - c_{kt\omega}^{food} - r_{kt\omega} - \lambda_{kt\omega}. \quad (14)$$

Every execution must preserve nonnegative inventory and dimensional consistency. A result that creates mass, erases rejected-batch inputs, or double-counts water is invalid.

Acceptability is not a cosmetic modifier. If $A_{jg} \in [0, 1]$ is a measured or locally reviewed acceptability coefficient and m_{jgtw} is the amount offered, then willing, safe uptake is bounded by

$$v_{jgtw} \leq A_{jg} m_{jgtw}. \quad (15)$$

This variable cannot authorize coercion. Religious practice, cultural meaning, preparation burden, communication, local participation, and the right to refuse remain governance constraints.

8. Robust and adaptive decision problem

First-stage decisions include prepositioning, contracts, qualified standby capacity, training, testing infrastructure, and initial allocation. Recourse decisions alter procurement, allocation, production scheduling, distribution, or substitution after observations arrive. A two-stage robust form is

$$\min_{x \in X} \left[c^T x + \max_{\omega \in \Omega} \min_{y(\omega) \in Y(x, \omega)} Q(x, y(\omega), \omega) \right]. \quad (16)$$

Cost cannot precede safety. The declared lexicographic objective is

$$Q = \text{lexmin}(V_{hard}, S_{worst}, S_{expected}, C_{fragile}, G_{inequity}, T_{recovery}, Cost). \quad (17)$$

Here $V_{hard} = 0$ is required for hard water, safety, clinical, and legal conditions. Weighted scenario shortfall and worst-case shortfall are

$$S_{\omega} = \sum_{g,n,t} \alpha_{gnt} s_{gnt\omega}, \quad S_{worst} = \max_{\omega \in \Omega} S_{\omega}. \quad (18)$$

Rolling-horizon revision is

$$\pi_{t+1} = \mathcal{R}(\pi_t, \mathcal{O}_{t+1}, \mathcal{E}_{t+1}), \quad (19)$$

where observations include demand, inventory, infrastructure, and production state, while evidence updates include qualification gates and validity windows. No update can retroactively release an unqualified batch.

Two scenarios with identical observed histories must receive the same action before they diverge:

$$\mathcal{H}_t(\omega) = \mathcal{H}_t(\omega') \implies y_t(\omega) = y_t(\omega'). \quad (20)$$

Every hard robust constraint is checked over its declared uncertainty set:

$$g_k(x, y(\omega), \theta) \leq 0 \quad \forall k, \forall \theta \in \mathcal{U}_{\omega}, \forall \omega \in \Omega. \quad (21)$$

The lexicographic objective is implemented as sequential constrained solves rather than an undocumented weighted sum. If $z_1, \dots, z_m$ are ordered components and earlier optima are certified, stage r is

$$z_r^* = \min z_r \quad \text{subject to model constraints and } z_i \leq z_i^* \ (i < r). \quad (22)$$

If a stage cannot be certified, the result is unresolved. A timeout is not infeasibility, and a weighted surrogate cannot silently replace the declared ordering.

9. Infeasibility and marginal value

Emergency software is often pressured to return an answer. This application must sometimes refuse. If the hard-gated feasible set is empty,

$$\mathcal{F}(\omega) = \emptyset, \quad (23)$$

the system returns an infeasibility certificate listing irreducible deficits, unavailable resources, failed or unknown gates, and affected populations. Relaxation scenarios may be shown, but each relaxation remains labeled and cannot be presented as safe compliance.

The marginal value of candidate module (j) is evaluated against a matched baseline:

$$\Delta_j = S_{worst}(baseline) - S_{worst}(baseline + j). \quad (24)$$

A positive value matters only if hard violations remain zero and no earlier objective worsens. This prevents high gross yield from appearing beneficial because the calculation omitted water, energy, testing, rejection, distribution, or clinical separation.

Recovery is also explicit. If $T_{j\omega}^{rec}$ is time to restore qualified operation, then

$$T_{recovery,\omega} = \min_{\pi \in \Pi_{safe}} \max_j T_{j\omega}^{rec}(\pi). \quad (25)$$

The safe-policy set preserves hard gates during transition.

10. Executable reference application

The reference application accepts a versioned scenario package, not free-text promises. It records population groups, planning requirements, starting inventory, verified suppliers, infrastructure states, uncertainty sets, module identity and preparation, lags and yields, composition with units [26], resource coefficients, evidence gates, test validity, jurisdiction, acceptability, dependencies, distribution capacity, and provenance.

It returns a feasible portfolio or an explicit refusal; nutrient coverage by group, time, nutrient, and scenario; resource use and rejected output; every failed or unknown gate; worst and expected shortfall; equity and concentration diagnostics; module marginal value and dependencies; fallback and recovery behavior; and a complete event and parameter log.

Five result classes remain distinct: invalid input, no solution proven, infeasible, feasible, and feasible only under an explicitly unsafe optimistic control. The application is blocked from emitting

household cultivation instructions, clinical recommendations, or deployment approval. It evaluates declared evidence; it does not manufacture evidence or authority.

11. Evaluation program

11.1 Development fixtures

The first executable study represented six policy conditions, fourteen disruption scenarios, eighteen ablations, sixteen edge fixtures, and eight independent counterexamples. It executed 84 baseline-scenario cells, 18 compound-scenario ablations, and 24 deterministic fixtures.

The 84 development cells contained 60 feasible safe-policy results, 10 infeasible safe-policy results, and 14 unsafe optimistic-control results. In two water-deficit scenarios, the optimistic control reported apparent feasibility while preserving the drinking-water violations that made the answer false-feasible. Seventeen of eighteen compound-scenario ablations remained infeasible. Only removal of the safe-water constraint converted the compound scenario to apparent feasibility.

The first tamper fixture passed 23 of 24 tests because it assigned 0.0 to a metric already equal to 0.0; no result byte changed, so the hash correctly remained valid. The fixture was repaired by changing the value. Independent verification then reported a result-hash mismatch and all 24 tests passed. The failed first attempt remains in the evidence history.

11.2 Isolated synthetic evaluation

A new 72-case partition was committed before outcome execution. It contained twelve cases in each of six synthetic families, three periods, four population types, nine nutrient dimensions, twelve dependency classes, and six matched policy conditions. Each condition received the same case data and 64-candidate mask budget, producing 432 saved results and 27,648 candidate evaluations. Exact paired sign tests received Holm familywise correction across four preregistered endpoints.

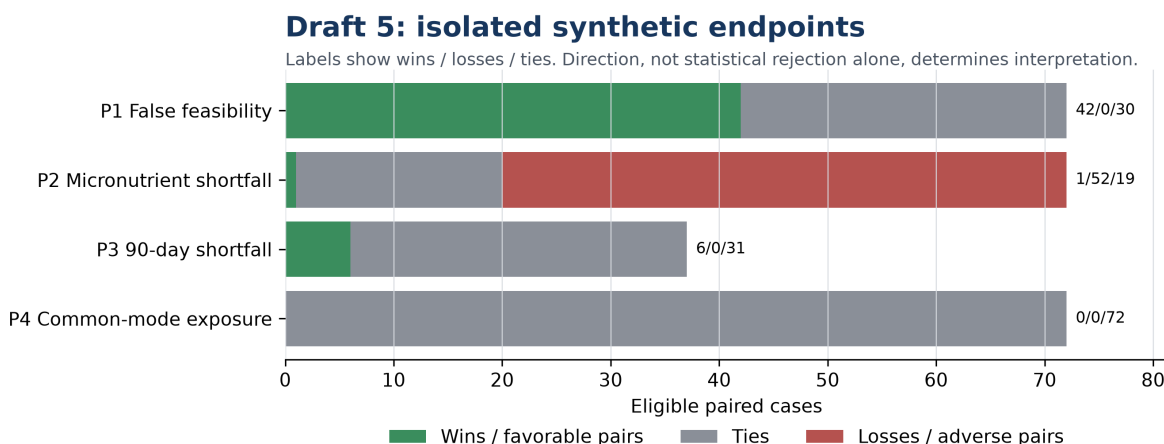


Figure 2. Draft 5 wins, losses, and ties. Lower-bound protection reduced nominal false feasibility, while the micronutrient hypothesis reversed, the production result did not survive multiplicity correction, and common-mode exposure was an exact tie.

Endpoint	Eligible	Wins	Losses	Ties	Holm p	Honest result
False feasibility: robust vs nominal	72	42	0	30	1.364e-12	favorable to uncertainty protection
Worst micronutrient shortfall	72	1	52	19	4.796e-14	adverse; hypothesis reversed
90-day shortfall vs conventional	37	6	0	31	0.0625	null after correction; tie-heavy
Common-mode exposure	72	0	0	72	1.0000	exact null

Robust MRFP, robust conventional, and simple fail-closed conditions each returned 34 feasible and 38 infeasible cases. Nominal MRFP and protein-only reported all 72 feasible, but lower-bound review found 42 and 41 false-feasible selections. The unsafe optimistic control labeled all 72 feasible under its unsafe rules and was false-feasible in 42. Thus false-feasibility protection was not unique to production or to the MRFP name; the strong conventional robust comparator achieved it too.

The micronutrient result is an important correction. A protein-only selector had lower worst micronutrient shortfall in 52 cases, while robust MRFP did so in one. This does not establish that protein-only planning is nutritionally adequate. It establishes that the first lexicographic complete-nutrient implementation failed its declared endpoint and required redesign.

11.3 Sensitivity, ablation, and scenario coupling

A separately frozen sensitivity study examined two nutrient-profile transformations, two placements of fragility, a three-level fractional grid, scenario-coupled uncertainty, branch-specific recourse, and production value against a coupled conventional baseline. It retained 121,608 candidate evaluations and reproduced exactly.

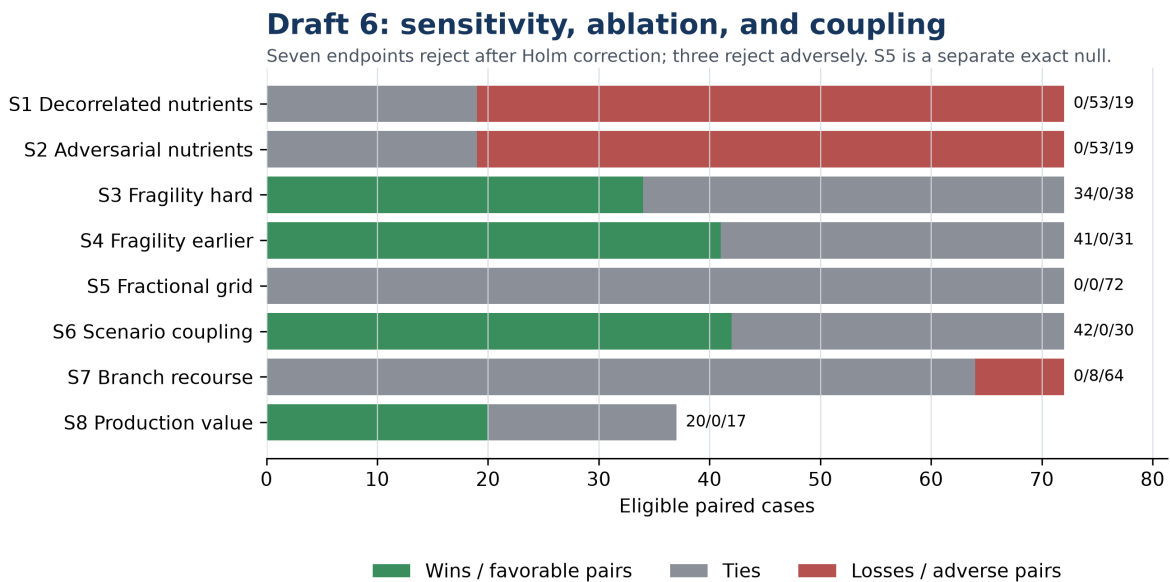


Figure 3. Draft 6 wins, losses, and ties. Seven endpoints rejected after Holm correction, and three of those rejected in an adverse direction. A separate eighth endpoint was an exact structural null. Rejection count is not success count.

Endpoint	Eligible	Wins	Losses	Ties	Holm p	Honest result
Decorrelated micronutrients	72	0	53	19	1.776e-15	adverse
Adversarial micronutrients	72	0	53	19	1.776e-15	adverse
Fragility hard constraint	72	34	0	38	4.657e-10	favorable but tie-heavy
Fragility earlier tier	72	41	0	31	4.547e-12	favorable
Fractional grid	72	0	0	72	1.0000	exact null
Scenario-coupled false feasibility	72	42	0	30	2.728e-12	favorable to coupling
Branch-specific recourse	72	0	8	64	0.015625	adverse formulation result
Coupled production value	37	20	0	17	5.722e-6	favorable in eligible subset

The adverse nutrient result survived both transformations. It was not explained by simple protein-micronutrient correlation. The correct conclusion remains that this objective formulation has not demonstrated superiority on its declared micronutrient endpoint.

Moving fragility earlier or making it a hard threshold changed decisions and lowered the modeled common-mode exposure. Scenario coupling removed 42 nominal false-feasible classifications. Yet the fractional grid tied the binary result in all 72 cases on worst shortfall, and branch-specific recourse lost in eight cases because it optimized local branch objectives rather than the same cross-branch maximum. That adverse result identified an objective mismatch, not a general claim that adaptation is harmful.

11.4 Solver reproducibility and independent route

The first mixed-integer study completed two 384-result CBC programs but differed at one case-condition identity. A separately frozen recovery study also failed exact replay when two different eligible identities timed out in opposite passes. A development-only reformulation was then aborted before partition outcomes because its verifier mishandled perfect-information production-energy accounting. These events remain preserved. Their outputs were not averaged, selected, repaired after viewing endpoints, or used for endpoint aggregation.

A new pre-outcome route used pinned single-threaded HiGHS 1.15.1, accepted only strict optimal status, serialized fresh case processes, saved process logs, and independently reconstructed constraints and metrics from the saved decisions. Three 96-result development passes reproduced exactly. Only then was a new 48-case scientific partition generated and committed. Two 384-result scientific passes reproduced exactly, all 768 results were strict optima, and every independent reconstruction passed.

The replay gates were

$$H(P_1) = H(P_2) = H(P_3), \quad |P_i| = 96, \quad (26)$$

for development and

$$H(S_1) = H(S_2), \quad |S_1| = |S_2| = 384. \quad (27)$$

for the scientific partition. Equality holds at the committed finite serialized-object boundary; it is not a proof of exact-real solver equivalence.

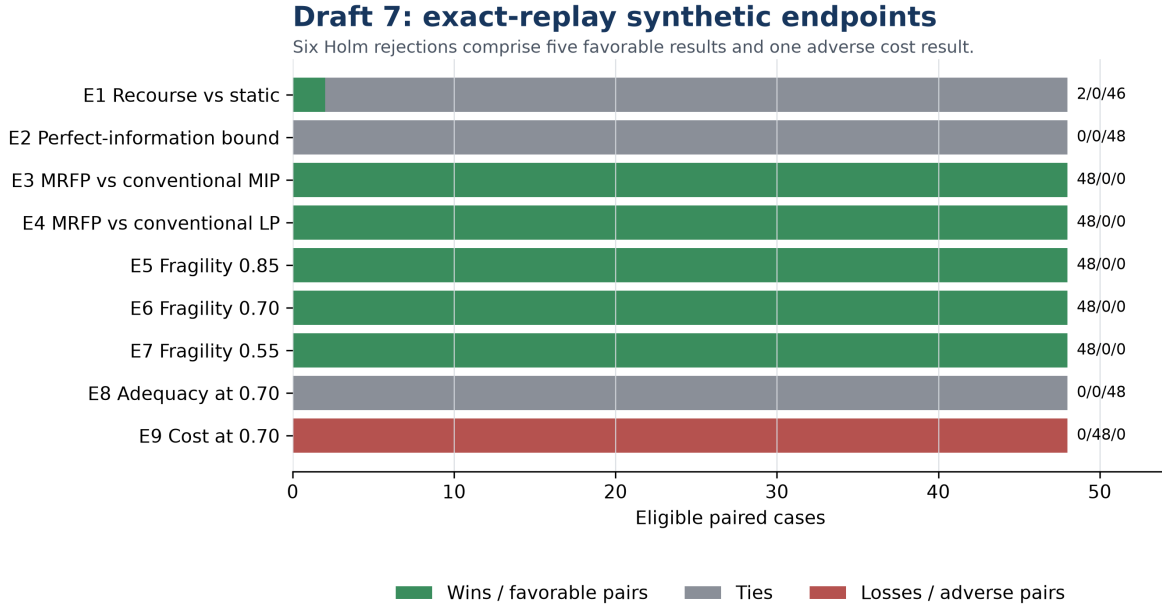


Figure 4. Draft 7 endpoint directions after the new replay gate passed. Two endpoints are exact paired nulls, the recourse result is favorable but tie-heavy, five comparisons are uniformly favorable, and the fragility-cost comparison is uniformly adverse.

Endpoint	Eligible	Wins	Losses	Ties	Median A-B	Holm p	Honest result
Recourse vs static shortfall	48	2	0	46	0.000000	1.0000	favorable but tie-heavy; null
Perfect-information bound	48	0	0	48	0.000000	1.0000	exact paired null
MRFP vs conventional MIP	48	48	0	0	-0.140067	6.395e-14	favorable in model
MRFP vs conventional LP	48	48	0	0	-0.140067	6.395e-14	favorable in model
Fragility threshold 0.85	48	48	0	0	-0.150000	6.395e-14	favorable exposure change

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Endpoint	Eligible	Wins	Losses	Ties	Median A-B	Holm p	Honest result
Fragility threshold 0.70	48	48	0	0	-0.300000	6.395e-14	favorable exposure change
Fragility threshold 0.55	48	48	0	0	-0.450000	6.395e-14	favorable exposure change
Threshold 0.70 adequacy	48	0	0	48	0.000000	1.0000	exact paired null
Threshold 0.70 cost	48	0	48	0	+111.464847	6.395e-14	adverse cost increase

Holm rejected six of nine endpoint nulls, but only five rejections favored the first named condition. The cost result rejected in the adverse direction. A count of statistical rejections is never reported as a count of successes.

12. Formal verification layer

The formal question is narrower than whether MRFP works in a real emergency: which evidence-discipline rules can be stated exactly and checked by a proof kernel?

Let endpoint eligibility ($E(s)$) equal one only for strict optimal solver status and zero for timeout, refusal, or process failure:

$$E(s) = \begin{cases} 1, & s = \textit{StrictOptimal}, \\ 0, & s \in \{\textit{Timeout}, \textit{Refusal}, \textit{ProcessFailure}\}. \end{cases} \quad (28)$$

For a pass (P) with expected size (n), strict-optimum count ($o(P)$), and retained failure count ($f(P)$), completeness is

$$C_n(P) \iff |P| = n \wedge o(P) = n \wedge f(P) = 0. \quad (29)$$

The scientific replay gate is

$$G_{384}(P_1, P_2) \iff C_{384}(P_1) \wedge C_{384}(P_2) \wedge H(P_1) = H(P_2). \quad (30)$$

For endpoint (e), wins, losses, and ties must exhaust eligible pairs:

$$w_e + \ell_e + t_e = n_e. \quad (31)$$

The Lean 4 leaf proves 29 bounded theorems: timeout, refusal, and process failure are endpoint-ineligible; incomplete or mismatched replay blocks aggregation; the nine endpoint count identities hold for the represented constants; direction follows wins versus losses rather than statistical significance; adding a passed study does not delete retained failure identifiers; an unopened final seal authorizes no payload access; unknown or rejected evidence maps to zero released quantity in the abstract model; and an inventory certificate satisfies its declared balance equation.

The shared compiler lease checked the source with Lean 4.30.0. The final check exited zero, contained no `sorry` or `admit`, and reported no axiom dependencies for the 29 named theorems. A deterministic bridge validator compares named JSON values and hashes with their Lean representation. That bridge is not a verified compiler or semantic refinement proof. The formal result is therefore a reporting and execution firewall, not a proof of empirical nutrition, real facility operation, exact-real solver optimality, or humanitarian effectiveness.

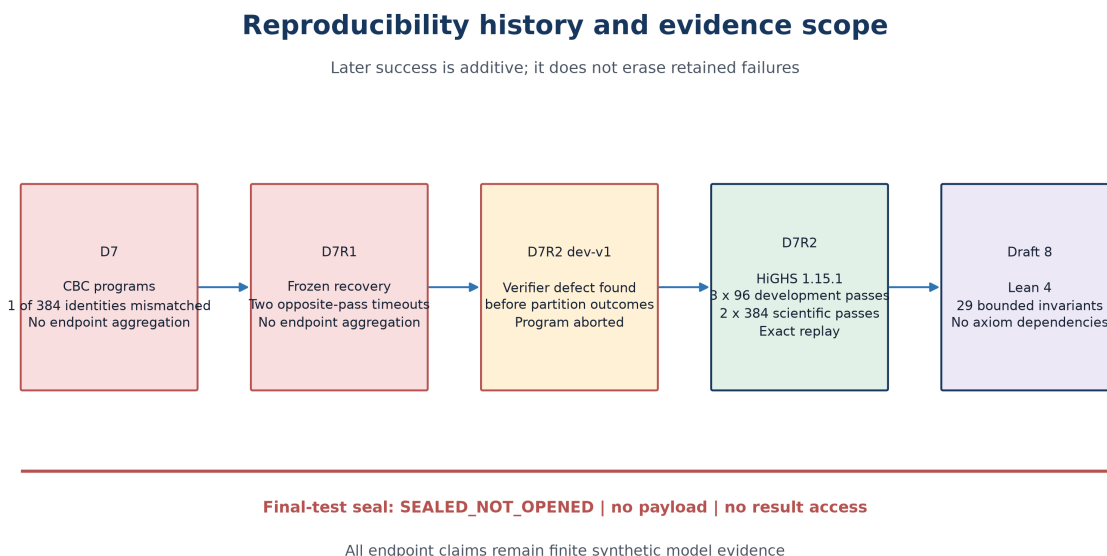


Figure 5. Evidence history is additive. The later exact HiGHS replay and Lean invariant layer do not erase the earlier CBC mismatch, opposite-pass timeouts, or pre-outcome verifier abort. The final-test seal remains closed.

13. Interpretation

The strongest complete interpretation contains both support and correction.

First, uncertainty protection and fail-closed release prevent a class of optimistic false-feasibility errors in the generated models. This mechanism is not unique to production-capable MRFP; a strong conventional robust comparator and a simple fail-closed policy also prevent those errors. MRFP is strongest as an integration framework that places later production beside, not above, established assistance.

Second, complete nutrition cannot be secured by merely naming more nutrients in an objective. The first implementation was worse than the protein-only comparator on its declared worst-micronutrient endpoint, and the reversal survived two transformed profiles. That failure does not rehabilitate protein as a sufficient nutrition target. It shows that the objective, action set, and allocation mechanics need to represent nutrient-wise tradeoffs more faithfully, potentially through Pareto, epsilon-constraint, or clinically governed adequacy structures.

Third, common-mode dependency matters only if it can affect selection. A late fragility term was inert. Promoting it changed portfolios, but stricter thresholds also increased modeled cost.

Resilience is not free, and the tradeoff must remain visible.

Fourth, recourse must be compared under the same aggregate objective. More flexible actions did not help when each branch minimized a different local tuple before a cross-branch maximum was reported. The adverse recourse result is a formulation diagnosis.

Fifth, exact replay is a prerequisite for endpoint interpretation, not a cosmetic quality metric. Two failed programs demonstrated that a completed solver study can remain scientifically unaggregable. The successful HiGHS route earned aggregation only after a development gate and two exact scientific passes.

Finally, synthetic precision cannot create empirical validity. Small p-values describe the frozen generated partitions. They do not estimate the safety of a food, the yield of a facility, the adequacy of a diet, the decisions of a regulator, the preferences of a community, or the effectiveness of humanitarian delivery.

14. Strongest alternatives and nonduplication boundary

The strongest established alternative is a nutritious conventional supply-chain optimizer with robust uncertainty protection, product qualification, rolling revision, and transparent infeasibility. The model must beat or complement this baseline, not a weakened caricature. In the first synthetic partition, the conventional robust comparator matched MRFP’s false-feasibility protection. In the later partition, MRFP improved modeled shortfall, but the generator and model construction still came from the same research program. Independent model and data authorship remain open.

A simpler fail-closed rule is another strong alternative. It matched the first robust MRFP portfolios exactly when fragility appeared too late to affect selection. That null directly motivated the later fragility tests.

The framework is not a claim that algae, insects, mushrooms, or microbial protein solve famine. Nor is it a recipe for cultivating them. Technology-specific evidence remains scoped to named species, preparations, products, processes, studies, facilities, populations, and jurisdictions.

The framework also does not duplicate a general theory of contractive computational dynamics. General computational work may contribute ancestry for convergence, mismatch, and failure-state concepts, but it lacks the food-specific population, nutrient, water, release, WASH, hazard, humanitarian, authority, acceptability, and recovery objects tested here.

The narrow novelty claim is therefore defensible: MRFP joins multi-clock assistance and production, typed qualification, complete resource and nutrient accounting, robust and adaptive comparison, explicit infeasibility, common-mode dependency, rights and authority, adverse-result retention, exact replay gates, and bounded formal evidence discipline in one food-resilience research program.

15. Ethics, authority, and deployment firewall

Food emergencies involve unequal power. A system that produces calories while coercing people, disrupting local markets, diverting drinking water, exposing communities to hazards, or substituting an experiment for humanitarian access has failed. Dignity, informed communication, cultural and religious choice, local participation, market impact, protection, environmental burden, labor safety,

and the needs of children, pregnant or lactating people, older adults, disabled people, and clinically malnourished people are substantive constraints.

No synthetic result authorizes real-world use. Field work would require competent local humanitarian, nutrition, WASH, food-safety, engineering, medical, legal, regulatory, and community governance. Each facility would need validated identity, inputs, process, operator training, monitoring, testing, release, traceability, corrective action, and recall. The package contains no household cultivation shortcuts and does not represent experimental biomass as treatment for malnutrition.

The final-test seal remains closed and contains no result payload. This manuscript does not infer a hidden final result from the seal. A local final-preprint candidate is not provider publication, peer review, field validation, or deployment authorization.

16. Falsification and next research gates

The contribution would narrow or fail if any of the following hold:

1. Explicit time horizons add no decision value beyond established nutritious supply-chain and rolling-horizon models.
2. Candidate production never improves a safely qualified medium- or long-horizon portfolio across defensible parameter ranges.
3. A simpler transparent rule matches the joined architecture across independently authored cases.
4. Conclusions depend on arbitrary objective order rather than stable constraint and sensitivity regions.
5. Fail-closed evidence cannot distinguish unknown, failed, expired, and not-applicable states audibly and reproducibly.
6. Units, preparations, populations, jurisdictions, or validity intervals cannot be normalized without hidden substitutions.
7. Infeasibility certificates cannot identify the real missing capacity or evidence.
8. Dependency and fragility controls merely transfer risk or cost without meaningful resilience.
9. The architecture diverts water, energy, labor, access, or funding from more effective humanitarian action.
10. Local rights and authority cannot be represented without implying that a global optimizer may decide for a community.

The next mathematical gate is a nutrient-wise Pareto or epsilon-constraint redesign, exact continuous or mixed-integer quantities with fixed and variable costs, and nonanticipative recourse under one shared aggregate objective. The next numerical gate is independent implementation with certificate-producing or exact-arithmetic checks where feasible. The next scientific gate is an independently authored generator or evidence-bound data set with domain review. The empirical gate requires food-specific composition, bioavailability, hazards, facility performance, regulation, logistics, acceptability, and field evidence. None is promoted by the present results.

17. Conclusion

Food resilience is simultaneously a timing, nutrition, water, safety, logistics, and governance problem. The fastest-growing organism is not necessarily the fastest safe meal. The largest protein output is not necessarily the best diet. A mathematically feasible result is not necessarily a valid decision.

MRFP supplies an accounting and refusal architecture for this problem. It begins with established food assistance and safe water, places candidate production on later and conditional clocks, charges every input and loss, separates growth from release and delivery, protects population-specific requirements, treats unknown mandatory evidence as closed, tests compound dependencies, and reports infeasibility or computational failure honestly.

The evaluation program strengthens and narrows that proposal. It demonstrates reproducible fail-closed model behavior and several favorable synthetic comparisons, but it also preserves a reversed nutrient hypothesis, exact nulls, tie-heavy results, adverse recourse and cost outcomes, two failed solver-replay programs, and a pre-outcome abort. The later successful replay and bounded Lean layer do not erase those facts.

The result is not a field-ready food system. It is a disciplined research framework for asking when an unconventional production module adds safe, complete, and resilient capacity beyond established humanitarian supply - and for refusing to say that it does when the evidence, model, computation, or authority is missing.

Data, code, and evidence availability

The local reproducibility companion contains the exact manuscripts, commitments, result receipts, application sources, frozen synthetic outputs, formal source and compiler receipts, claim and challenge ledgers, figures, metadata draft, validation report, and SHA-256 manifest used for this preprint candidate. The companion is a local review artifact and has not been uploaded or published. The final-test seal remains closed.

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